

6. Normal Distribution — Solved Problems in Data Science

1. What is Normal Distribution?

A **Normal Distribution** is a continuous probability distribution that is:

- Bell-shaped
- Symmetric around its mean
- μ
- σ

We write:

$$X \sim N(\mu, \sigma^2)$$

where:

- μ = mean
- σ = standard deviation
- σ^2 = variance

Probability Density Function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The PDF tells us the **shape/density** of the distribution. Probability over an interval is represented by the **area under the curve**.

2. Z-Score

The most important calculation with a Normal distribution is the **Z-score**.

$$Z = \frac{X - \mu}{\sigma}$$

A Z-score tells us **how many standard deviations an observation is away from the mean**.

For example:

$$Z = 2$$

means the observation is **2 standard deviations above the mean**.

Problem 1: Student Exam Scores

Suppose student scores in a large dataset are normally distributed with:

$$\begin{aligned}\mu &= 70 \\ \sigma &= 10\end{aligned}$$

Find the Z-score of a student who scored **85**.

Solution

Given:

$$X = 85, \quad \mu = 70, \quad \sigma = 10$$

Using:

$$\begin{aligned}Z &= \frac{X - \mu}{\sigma} \\ Z &= \frac{85 - 70}{10} \\ Z &= \frac{15}{10}\end{aligned}$$

$$Z = 1.5$$

Interpretation

The student's score is:

1.5 standard deviations above the mean

This is useful in Data Science because standardized scores allow us to compare observations measured on different scales.

Problem 2: Probability of a Score Below 80

Student scores are normally distributed:

$$X \sim N(70, 10^2)$$

Find:

$$P(X < 80)$$

Step 1: Calculate the Z-score

$$Z = \frac{80 - 70}{10}$$

$$Z = 1$$

From the standard Normal distribution table:

$$P(Z < 1) = 0.8413$$

Therefore:

$$P(X < 80) = 0.8413$$

Answer

$$84.13\%$$

Data Science Interpretation

Approximately **84.13% of students are expected to score below 80.**

Problem 3: Probability of a Score Above 80

Using the same distribution:

$$X \sim N(70, 10^2)$$

Find:

$$P(X > 80)$$

Solution

We already calculated:

$$Z = 1$$

From the Z-table:

$$P(Z < 1) = 0.8413$$

Therefore:

$$P(Z > 1) = 1 - 0.8413$$

$$P(Z > 1) = 0.1587$$

Answer

$$15.87\%$$

Interpretation

Approximately **15.87% of students are expected to score above 80.**

Problem 4: Probability Between Two Scores

Exam scores are normally distributed with:

$$\mu = 70$$

$$\sigma = 10$$

Find the probability that a randomly selected student scores between **60 and 80**.

Step 1: Convert 60 to Z-score

$$Z_1 = \frac{60 - 70}{10}$$

$$Z_1 = -1$$

Step 2: Convert 80 to Z-score

$$Z_2 = \frac{80 - 70}{10}$$

$$Z_2 = 1$$

Therefore:

$$P(60 < X < 80) = P(-1 < Z < 1)$$

From the standard Normal distribution:

$$P(-1 < Z < 1) \approx 0.6827$$

Answer

68.27%

Important Rule

Approximately:

68.27%

of observations lie within **1 standard deviation** of the mean for a Normal distribution.

Problem 5: Employee Salary Analysis

Suppose salaries in a particular job category are approximately normally distributed with:

$$\mu = ₹50,000$$

and

$$\sigma = ₹8,000$$

Find the Z-score for an employee earning **₹66,000**.

Solution

$$\begin{aligned} Z &= \frac{X - \mu}{\sigma} \\ Z &= \frac{66000 - 50000}{8000} \\ Z &= \frac{16000}{8000} \end{aligned}$$

$$Z = 2$$

Interpretation

The employee's salary is:

2 standard deviations above the mean

This can be useful in **salary analytics and anomaly detection**.

Problem 6: Probability of Salary Above ₹66,000

Using the same salary distribution:

$$\begin{aligned}\mu &= ₹50,000 \\ \sigma &= ₹8,000\end{aligned}$$

Find the probability that an employee earns more than ₹66,000.

Step 1: Calculate Z

$$\begin{aligned}Z &= \frac{66000 - 50000}{8000} \\ Z &= 2\end{aligned}$$

From the Z-table:

$$P(Z < 2) = 0.9772$$

Therefore:

$$P(Z > 2) = 1 - 0.9772$$

$$P(Z > 2) = 0.0228$$

Answer

2.28%

Interpretation

Only approximately **2.28% of employees** are expected to earn more than ₹66,000 under this Normal model.

Problem 7: Website Response Time

Suppose API response times are approximately normally distributed:

$$\begin{aligned}\mu &= 200 \text{ ms} \\ \sigma &= 40 \text{ ms}\end{aligned}$$

Find the probability that a response takes **less than 240 ms**.

Solution

Calculate Z:

$$Z = \frac{240 - 200}{40}$$

$$Z = 1$$

Therefore:

$$P(X < 240) = P(Z < 1)$$

$$P(X < 240) = 0.8413$$

Answer

84.13%

Data Science Interpretation

Approximately **84.13% of API requests are expected to complete within 240 ms.**

This can be useful for monitoring **service-level performance.**

Problem 8: Detecting an Unusually Slow Response

Using the same API data:

$$\begin{aligned}\mu &= 200 \text{ ms} \\ \sigma &= 40 \text{ ms}\end{aligned}$$

A request takes **320 ms.**

Determine its Z-score and whether it appears unusual under a simple Normal-model rule.

Solution

$$Z = \frac{320 - 200}{40}$$

$$Z = \frac{120}{40}$$

$$Z = 3$$

The response time is **3 standard deviations above the mean**.

For a Normal distribution, observations beyond approximately:

$$\mu \pm 3\sigma$$

are relatively rare.

Answer

$$Z = 3$$

The observation would generally be considered unusually high under a simple 3σ screening rule.

Important Data Science Note

A high Z-score does **not automatically mean an observation is an error**. It may represent a genuine unusual event.

Problem 9: Machine Sensor Data

A temperature sensor produces readings that are approximately normally distributed with:

$$\begin{aligned}\mu &= 50^\circ C \\ \sigma &= 5^\circ C\end{aligned}$$

Find the probability that the temperature is between:

$$45^\circ C$$

and

$$55^\circ C$$

Solution

For $45^\circ C$:

$$\begin{aligned}Z_1 &= \frac{45 - 50}{5} \\ Z_1 &= -1\end{aligned}$$

For $55^\circ C$:

$$Z_2 = \frac{55 - 50}{5}$$

$$Z_2 = 1$$

Therefore:

$$P(45 < X < 55) = P(-1 < Z < 1)$$

$$P(-1 < Z < 1) = 0.6827$$

Answer

$$68.27\%$$

Problem 10: Finding a Percentile

Suppose customer spending is normally distributed with:

$$\mu = ₹2,000$$

$$\sigma = ₹500$$

Find the spending amount corresponding to approximately the **95th percentile**.

For the 95th percentile:

$$Z \approx 1.645$$

We use:

$$Z = \frac{X - \mu}{\sigma}$$

Rearranging:

$$X = \mu + Z\sigma$$

Therefore:

$$X = 2000 + (1.645)(500)$$

$$X = 2000 + 822.5$$

$$X = ₹2,822.50$$

Answer

The 95th percentile is approximately:

₹2,822.50

Interpretation

Approximately **95% of customers are expected to spend less than ₹2,822.50**, assuming the Normal model is appropriate.

Problem 11: Finding the Value Corresponding to the 90th Percentile

Suppose:

$$X \sim N(100, 15^2)$$

Find the value corresponding to the **90th percentile**.

For the 90th percentile:

$$Z \approx 1.2816$$

Using:

$$\begin{aligned} X &= \mu + Z\sigma \\ X &= 100 + (1.2816)(15) \\ X &= 100 + 19.224 \end{aligned}$$

$$X \approx 119.22$$

Answer

119.22

Problem 12: Finding the Probability Within Two Standard Deviations

Suppose:

$$X \sim N(100, 15^2)$$

Find the probability that X lies between:

70

and

130

Step 1: Calculate lower Z-score

$$Z_1 = \frac{70 - 100}{15}$$

$$Z_1 = -2$$

Step 2: Calculate upper Z-score

$$Z_2 = \frac{130 - 100}{15}$$

$$Z_2 = 2$$

Therefore:

$$P(70 < X < 130) = P(-2 < Z < 2)$$

Using the standard Normal distribution:

$$P(-2 < Z < 2) \approx 0.9545$$

Answer

95.45%

Problem 13: Data Science — Standardizing a Dataset

A dataset contains an age feature with:

$$\mu = 40$$

$$\sigma = 8$$

A person's age is:

$$X = 56$$

Find the standardized value.

Solution

$$\begin{aligned}Z &= \frac{X - \mu}{\sigma} \\Z &= \frac{56 - 40}{8} \\Z &= \frac{16}{8} \\Z &= 2\end{aligned}$$

Interpretation

The person's age is **2 standard deviations above the dataset mean**.

Why is this useful?

Z-score standardization is widely used before applying algorithms where feature scale matters.

For a dataset, each value can be transformed as:

$$X_{\text{standardized}} = \frac{X - \mu}{\sigma}$$

After standardization, the transformed feature has approximately:

$$\mu = 0$$

and

$$\sigma = 1$$

when the mean and standard deviation used are those of the dataset.

Problem 14: Comparing Two Students

Student A scored **80** in Mathematics, where:

$$\mu = 70, \quad \sigma = 5$$

Student B scored **85** in Statistics, where:

$$\mu = 75, \quad \sigma = 8$$

Who performed better **relative to their respective classes**?

Student A

$$Z_A = \frac{80 - 70}{5}$$

$$Z_A = 2$$

Student B

$$Z_B = \frac{85 - 75}{\frac{8}{10}}$$

$$Z_B = 1.25$$

Comparison

$$\begin{aligned} Z_A &= 2 \\ Z_B &= 1.25 \end{aligned}$$

Therefore:

Student A performed better relative to their class

Data Science Application

This is exactly why **standardization** is useful.

Raw scores cannot always be compared directly when the underlying distributions have different means and standard deviations.

Problem 15: Real Data Science Problem — Anomaly Detection

A company's transaction amounts are approximately normally distributed:

$$\begin{aligned} \mu &= ₹1,000 \\ \sigma &= ₹200 \end{aligned}$$

A new transaction has an amount of:

₹1,800

Determine whether it is unusual using a simple **3-standard-deviation rule**.

Solution

Calculate the Z-score:

$$Z = \frac{1800 - 1000}{\frac{200}{800}}$$
$$Z = \frac{800}{200}$$

$$Z = 4$$

The transaction is:

4 standard deviations above the mean

Under a Normal model, such an observation is extremely unusual.

Approximate probability

For a standard Normal variable:

$$P(Z > 4) \approx 0.0000317$$

Therefore:

$$P(Z > 4) \approx 0.00317\%$$

Data Science Interpretation

The transaction is a strong **candidate for anomaly investigation**.

However, a Data Scientist should **not automatically classify it as fraud**.

The system could investigate:

- Customer history
- Location
- Device information
- Transaction frequency
- Previous spending patterns
- Merchant information
- Authentication details

This illustrates an important principle:

Anomaly $\neq$ Fraud

An anomaly is an observation that is unusual according to the chosen statistical model.

Problem 16: Six-Sigma Style Quality Control

A manufacturing process produces components with:

$$\mu = 100 \text{ mm}$$

$$\sigma = 2 \text{ mm}$$

Find the range corresponding to ± 3 **standard deviations** from the mean.

Solution

Lower limit:

$$\begin{aligned} \mu - 3\sigma \\ \mu - 3\sigma = 100 - 3(2) \end{aligned}$$

$$94 \text{ mm}$$

Upper limit:

$$\begin{aligned} \mu + 3\sigma \\ \mu + 3\sigma = 100 + 3(2) \end{aligned}$$

$$106 \text{ mm}$$

Therefore:

$$94 \leq X \leq 106$$

contains approximately:

$$99.73\%$$

of observations under a Normal model.

Data Science Application

This concept is useful in:

- Quality control
- Manufacturing analytics

- Sensor monitoring
 - Process monitoring
 - Outlier detection
-

Problem 17: Probability of an Extreme Observation

Suppose:

$$X \sim N(50, 10^2)$$

Find the probability that:

$$X > 70$$

Solution

Calculate the Z-score:

$$Z = \frac{70 - 50}{10}$$

$$Z = 2$$

From the standard Normal table:

$$P(Z < 2) = 0.9772$$

Therefore:

$$P(Z > 2) = 1 - 0.9772$$

$$P(X > 70) = 0.0228$$

Answer

$$2.28\%$$

Problem 18: Probability Between 40 and 60

Suppose:

$$X \sim N(50, 10^2)$$

Find:

$$P(40 < X < 60)$$

Solution

For $X = 40$:

$$Z_1 = \frac{40 - 50}{10} = -1$$

For $X = 60$:

$$Z_2 = \frac{60 - 50}{10} = 1$$

Therefore:

$$P(40 < X < 60) = P(-1 < Z < 1)$$

$$P = 0.6827$$

Answer

$$68.27\%$$



Empirical Rule — Very Important

For a Normal distribution:

Within 1 standard deviation

$$\mu - \sigma < X < \mu + \sigma$$

Approximately:

$$68.27\%$$

Within 2 standard deviations

$$\mu - 2\sigma < X < \mu + 2\sigma$$

Approximately:

95.45%

Within 3 standard deviations

$$\mu - 3\sigma < X < \mu + 3\sigma$$

Approximately:

99.73%

Easy memory trick

68 – 95 – 99.7

This is called the **Empirical Rule** or **68–95–99.7 Rule**.

Important Normal Distribution Formulas

Normal Distribution

$$X \sim N(\mu, \sigma^2)$$

PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Z-score

$$Z = \frac{X - \mu}{\sigma}$$

Converting Z-score back to X

$$X = \mu + Z\sigma$$

Mean

$$E(X) = \mu$$

Variance

$$Var(X) = \sigma^2$$

Standard deviation

$$\sigma = \sqrt{Var(X)}$$



Normal Distribution in Data Science

Application	Example
Feature Standardization	Convert age, income, height to Z-scores
Outlier Detection	Identify observations with large (
Anomaly Detection	Detect unusually high/low transactions
Quality Control	Monitor manufacturing measurements
Sensor Analytics	Identify unusual sensor readings
Performance Analysis	Compare observations relative to a population
Statistical Inference	Confidence intervals and hypothesis testing
A/B Testing	Approximate sampling distributions in suitable settings
Data Visualization	Understand bell-shaped distributions
Machine Learning	Gaussian assumptions in some models and transformations



Comparing the Distributions You Have Learned

Distribution	Main Question	Example
Bernoulli	What happens in one trial?	Does a customer purchase?
Binomial	How many successes in fixed trials?	How many customers purchase among 100?
Geometric	How many trials until first success?	How many visitors until first purchase?
Poisson	How many events in an interval?	How many requests per second?
Uniform	Are values equally distributed over an interval?	Random value between 0 and 1

Distribution	Main Question	Example
Normal	How are continuous values distributed around a mean?	Exam scores, measurement errors



One-Line Intuition

Remember Normal Distribution like this:

"Most observations are close to the mean, and observations become increasingly rare as we move farther away from the mean."

For example:

$$\mu = 100, \quad \sigma = 10$$

Then:

$$\begin{aligned} 90 &= \mu - 1\sigma \\ 110 &= \mu + 1\sigma \end{aligned}$$

Approximately **68.27%** of observations fall between 90 and 110.

Similarly:

$$\begin{aligned} 80 &= \mu - 2\sigma \\ 120 &= \mu + 2\sigma \end{aligned}$$

Approximately **95.45%** fall between 80 and 120.

And:

$$\begin{aligned} 70 &= \mu - 3\sigma \\ 130 &= \mu + 3\sigma \end{aligned}$$

Approximately **99.73%** fall between 70 and 130.

The key idea for Data Science:

Raw value → Z – score → Probability / Percentile / Anomaly decision

That **Z-score** → **probability** → **interpretation** workflow is one of the most important practical uses of the Normal distribution in Data Science.

give solved problems on **Exponential Distribution** in the context of data science

Absolutely 👍 Let's continue the series. **Exponential Distribution** is especially useful in Data Science when we are interested in **waiting time until the next event**—for example, the time until the next customer arrives, server failure, API request, or machine breakdown.

7. Exponential Distribution — Solved Problems in Data Science

1. What is Exponential Distribution?

The **Exponential Distribution** is a continuous probability distribution commonly used to model the **waiting time until the next event occurs**.

If the event rate is λ , we write:

$$X \sim \text{Exponential}(\lambda)$$

where:

- X = waiting time
- λ = average event rate
- $e \approx 2.71828$

The **Probability Density Function (PDF)** is:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

and:

$$f(x) = 0, \quad x < 0$$

2. Important Formulas

Probability that waiting time is less than x

The Cumulative Distribution Function (CDF) is:

$$P(X \leq x) = 1 - e^{-\lambda x}$$

Probability that waiting time is greater than x

$$P(X > x) = e^{-\lambda x}$$

Probability between two values

$$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$$

Mean

$$E(X) = \frac{1}{\lambda}$$

Variance

$$Var(X) = \frac{1}{\lambda^2}$$

Standard deviation

$$\sigma = \frac{1}{\lambda}$$

So, for an Exponential distribution:

$$\text{Mean} = \text{Standard Deviation} = \frac{1}{\lambda}$$

Problem 1: Website Visitor Arrival

A website receives an average of **6 visitors per minute**.

Assume the time between consecutive visitors follows an Exponential distribution.

Find the probability that the next visitor arrives within **10 seconds**.

Step 1: Convert the time

Since the rate is given per minute:

$$10 \text{ seconds} = \frac{10}{60} = \frac{1}{6} \text{ minute}$$

Therefore:

$$x = \frac{1}{6}$$

and:

$$\lambda = 6$$

Step 2: Use the CDF

$$P(X \leq x) = 1 - e^{-\lambda x}$$

Therefore:

$$P(X \leq \frac{1}{6}) = 1 - e^{-6(\frac{1}{6})}$$
$$P(X \leq \frac{1}{6}) = 1 - e^{-1}$$

Since:

$$e^{-1} \approx 0.3679$$

we get:

$$P(X \leq \frac{1}{6}) = 1 - 0.3679$$

$$P(X \leq 10 \text{ sec}) \approx 0.6321$$

Answer

$$63.21\%$$

Data Science Interpretation

There is approximately a **63.21% probability that the next visitor arrives within 10 seconds.**

Problem 2: API Request Waiting Time

An API receives an average of **12 requests per second.**

What is the probability that the next request arrives within **0.1 seconds?**

Given

$$\lambda = 12$$

$$x = 0.1$$

Using:

$$P(X \leq x) = 1 - e^{-\lambda x}$$

we get:

$$\begin{aligned} P(X \leq 0.1) &= 1 - e^{-12(0.1)} \\ P(X \leq 0.1) &= 1 - e^{-1.2} \end{aligned}$$

Since:

$$e^{-1.2} \approx 0.3010$$

therefore:

$$P(X \leq 0.1) = 1 - 0.3010$$

$$P(X \leq 0.1) \approx 0.6990$$

Answer

69.90%

Interpretation

There is approximately a **69.90% probability that the next API request arrives within 0.1 seconds.**

Problem 3: Waiting More Than 30 Seconds

A customer-support system receives an average of **4 calls per minute**.

Find the probability that the waiting time for the next call is **more than 30 seconds**.

Step 1: Convert time

$$30 \text{ seconds} = 0.5 \text{ minute}$$

Therefore:

$$x = 0.5$$

and:

$$\lambda = 4$$

Step 2: Use survival probability

$$P(X > x) = e^{-\lambda x}$$

Therefore:

$$P(X > 0.5) = e^{-4(0.5)}$$

$$P(X > 0.5) = e^{-2}$$

$$P(X > 0.5) \approx 0.1353$$

Answer

$$13.53\%$$

Interpretation

There is approximately a **13.53% chance that the support center waits more than 30 seconds for the next call.**

Problem 4: Expected Waiting Time for a Customer

A website receives an average of **5 customers per minute**.

Find the expected waiting time until the next customer arrives.

Solution

Given:

$$\lambda = 5 \text{ customers/minute}$$

The expected waiting time is:

$$E(X) = \frac{1}{\lambda}$$

Therefore:

$$E(X) = \frac{1}{5}$$

$$E(X) = 0.2 \text{ minutes}$$

Convert to seconds:

$$0.2 \times 60 = 12$$

Answer

12 seconds

Interpretation

On average, the website waits approximately **12 seconds between customer arrivals**.

Problem 5: Server Failure Analysis

A server experiences an average of **2 failures per month**.

Assume the time until the next failure follows an Exponential distribution.

Find the probability that the server operates for **at least 3 months without failure**.

Given

$$\begin{aligned}\lambda &= 2 \text{ failures/month} \\ x &= 3 \text{ months}\end{aligned}$$

We need:

$$P(X > 3)$$

Using:

$$P(X > x) = e^{-\lambda x}$$

we get:

$$\begin{aligned}P(X > 3) &= e^{-2(3)} \\ P(X > 3) &= e^{-6}\end{aligned}$$

$$P(X > 3) \approx 0.00248$$

Answer

0.248%

Interpretation

There is approximately a **0.248% probability that the server operates for more than 3 months without failure**, under this model.

Problem 6: Machine Breakdown

A manufacturing machine has an average failure rate of:

$$0.05 \text{ failures/hour}$$

Find the probability that the machine does **not fail during the next 10 hours**.

Solution

Given:

$$\begin{aligned}\lambda &= 0.05 \\ x &= 10\end{aligned}$$

We need:

$$P(X > 10)$$

Therefore:

$$\begin{aligned}P(X > 10) &= e^{-0.05(10)} \\ P(X > 10) &= e^{-0.5}\end{aligned}$$

$$P(X > 10) \approx 0.6065$$

Answer

$$60.65\%$$

Interpretation

There is approximately a **60.65% probability that the machine survives the next 10 hours without failure**.

Problem 7: Probability of a Waiting Time Between 2 and 5 Minutes

A call center receives an average of **3 calls per minute**.

Find the probability that the waiting time until the next call is between **2 and 5 minutes**.

Given

$$\lambda = 3$$

We need:

$$P(2 < X < 5)$$

Using:

$$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$$

Therefore:

$$\begin{aligned} P(2 < X < 5) &= e^{-3(2)} - e^{-3(5)} \\ P(2 < X < 5) &= e^{-6} - e^{-15} \end{aligned}$$

Using:

$$e^{-6} \approx 0.002479$$

and:

$$e^{-15} \approx 0.000000306$$

Therefore:

$$P(2 < X < 5) \approx 0.0024787$$

Answer

0.2479%

Interpretation

There is only about a **0.248% probability that the waiting time is between 2 and 5 minutes**.

This makes sense because the average waiting time is only:

$$\frac{1}{3} \text{ minute} = 20 \text{ seconds}$$

Problem 8: Finding the Rate from Mean Waiting Time

A Data Scientist observes that the average waiting time between API requests is **0.25 seconds**.

Find the corresponding Exponential event rate.

Given

$$E(X) = 0.25$$

For an Exponential distribution:

$$E(X) = \frac{1}{\lambda}$$

Therefore:

$$0.25 = \frac{1}{\lambda}$$

Rearranging:

$$\lambda = \frac{1}{0.25}$$

$$\lambda = 4$$

Answer

4 requests per second

Data Science Interpretation

If the average inter-arrival time is **0.25 seconds**, the estimated event rate is **4 requests per second**.

Problem 9: Mean and Variance of Waiting Time

An online system receives an average of **10 requests per second**.

Find:

1. Mean waiting time
2. Variance
3. Standard deviation

Given

$$\lambda = 10$$

Mean

$$E(X) = \frac{1}{\lambda}$$

$$E(X) = 0.1 \text{ seconds}$$

Variance

$$Var(X) = \frac{1}{\lambda^2}$$

$$Var(X) = \frac{1}{10^2}$$

$$Var(X) = 0.01 \text{ seconds}^2$$

Standard deviation

$$\sigma = \frac{1}{\lambda}$$

$$\sigma = 0.1 \text{ seconds}$$

Answer

Measure	Value
Mean	0.1 sec
Variance	0.01 sec ²
Standard deviation	0.1 sec

Problem 10: Customer Arrival — Within One Minute

A store receives customers at an average rate of:

3 customers/minute

Find the probability that the next customer arrives within **1 minute**.

Solution

Given:

$$\begin{aligned}\lambda &= 3 \\ x &= 1\end{aligned}$$

Using:

$$\begin{aligned}P(X \leq x) &= 1 - e^{-\lambda x} \\ P(X \leq 1) &= 1 - e^{-3}\end{aligned}$$

Since:

$$e^{-3} \approx 0.0498$$

we get:

$$P(X \leq 1) = 1 - 0.0498$$

$$P(X \leq 1) \approx 0.9502$$

Answer

$$95.02\%$$

Interpretation

There is approximately a **95.02% probability that the next customer arrives within one minute**.

Problem 11: Finding the Median Waiting Time

An API receives requests at a rate of:

$$\lambda = 2 \text{ requests/second}$$

Find the **median waiting time**.

Solution

The median m satisfies:

$$P(X \leq m) = 0.5$$

For an Exponential distribution:

$$1 - e^{-\lambda m} = 0.5$$

Therefore:

$$e^{-\lambda m} = 0.5$$

Taking natural logarithms:

$$-\lambda m = \ln(0.5)$$

Since:

$$\ln(0.5) = -\ln 2$$

we get:

$$m = \frac{\ln 2}{\lambda}$$

Therefore:

$$m = \frac{0.6931}{2}$$

$$m \approx 0.3466 \text{ seconds}$$

Answer

$$0.347 \text{ seconds approximately}$$

Important Observation

The mean waiting time is:

$$\frac{1}{2} = 0.5 \text{ seconds}$$

but the median is:

$$0.347 \text{ seconds}$$

This happens because the Exponential distribution is **right-skewed**.

Problem 12: 95th Percentile of Waiting Time

An API receives requests at an average rate of:

$$\lambda = 5 \text{ requests/second}$$

Find the **95th percentile** of waiting time.

Solution

The 95th percentile x satisfies:

$$P(X \leq x) = 0.95$$

Using the Exponential CDF:

$$1 - e^{-5x} = 0.95$$

Therefore:

$$e^{-5x} = 0.05$$

Taking natural logarithms:

$$-5x = \ln(0.05)$$

Thus:

$$x = -\frac{\ln(0.05)}{5}$$

Since:

$$-\ln(0.05) \approx 2.9957$$

we get:

$$x = \frac{2.9957}{5}$$

$$x \approx 0.5991 \text{ seconds}$$

Answer

$$95^{th} \text{ percentile} \approx 0.599 \text{ seconds}$$

Interpretation

Approximately **95% of waiting times are expected to be less than 0.599 seconds** under the model.

Problem 13: Reliability Analysis

A component has an average lifetime of **1000 hours**.

Assume its lifetime follows an Exponential distribution.

Find the probability that the component lasts **more than 1500 hours**.

Step 1: Find λ

The mean is:

$$E(X) = \frac{1}{\lambda}$$

Given:

$$E(X) = 1000$$

Therefore:

$$\lambda = \frac{1}{1000}$$

Step 2: Calculate survival probability

We need:

$$P(X > 1500)$$

Using:

$$\begin{aligned} P(X > x) &= e^{-\lambda x} \\ P(X > 1500) &= e^{-\frac{1}{1000}(1500)} \\ P(X > 1500) &= e^{-1.5} \end{aligned}$$

$$P(X > 1500) \approx 0.2231$$

Answer

22.31%

Data Science Application

This type of calculation is used in:

- Reliability analytics
 - Predictive maintenance
 - Equipment lifetime modeling
 - Failure-time analysis
-

Problem 14: Predictive Maintenance

A machine fails on average **once every 20 days**.

Find the probability that it will operate for at least **30 more days** without failure.

Step 1: Determine rate

Mean lifetime:

$$E(X) = 20 \text{ days}$$

Therefore:

$$\lambda = \frac{1}{20} = 0.05$$

Step 2: Calculate probability

$$P(X > 30) = e^{-0.05(30)}$$

$$P(X > 30) = e^{-1.5}$$

$$P(X > 30) \approx 0.2231$$

Answer

22.31%

Interpretation

There is approximately a **22.31% probability that the machine operates for another 30 days without failure**, assuming the Exponential model.

Problem 15: Real Data Science Problem — Website Traffic

A website receives an average of **120 visitors per hour**.

A Data Scientist wants to analyze the time between successive visitors.

Find:

1. The event rate per minute
2. The average waiting time
3. The probability that the next visitor arrives within 15 seconds
4. The probability that the waiting time exceeds 30 seconds

Solution

Step 1: Convert the rate

Given:

120 visitors/hour

There are 60 minutes in an hour.

Therefore:

$$\lambda = \frac{120}{60}$$

$$\lambda = 2 \text{ visitors/minute}$$

Step 2: Average waiting time

For Exponential distribution:

$$E(X) = \frac{1}{\lambda}$$

Therefore:

$$E(X) = \frac{1}{2}$$
$$E(X) = 0.5 \text{ minute}$$

Convert to seconds:

$$0.5 \times 60 = 30$$

Therefore:

$$E(X) = 30 \text{ seconds}$$

Step 3: Probability next visitor arrives within 15 seconds

Convert:

$$15 \text{ seconds} = 0.25 \text{ minute}$$

We need:

$$P(X \leq 0.25)$$

Using:

$$\begin{aligned} P(X \leq x) &= 1 - e^{-\lambda x} \\ P(X \leq 0.25) &= 1 - e^{-2(0.25)} \\ P(X \leq 0.25) &= 1 - e^{-0.5} \\ P(X \leq 0.25) &= 1 - 0.6065 \end{aligned}$$

$$P(X \leq 0.25) \approx 0.3935$$

Therefore:

$$39.35\%$$

Step 4: Probability waiting exceeds 30 seconds

Convert:

$$30 \text{ seconds} = 0.5 \text{ minute}$$

We need:

$$P(X > 0.5)$$

Therefore:

$$\begin{aligned} P(X > 0.5) &= e^{-2(0.5)} \\ P(X > 0.5) &= e^{-1} \end{aligned}$$

$$P(X > 0.5) \approx 0.3679$$

Therefore:

$$36.79\%$$

Final Answer

Quantity	Result
Arrival rate	2 visitors/minute
Average waiting time	30 seconds
$P(X \leq 15 \text{ sec})$	39.35%
$P(X > 30 \text{ sec})$	36.79%

Data Science Interpretation

A Data Scientist can use this model to understand **inter-arrival times** and support:

- Website capacity planning
- Traffic modeling
- Queue analysis
- Server scaling
- Customer experience analysis

Problem 16: Important Connection — Poisson and Exponential

A server receives an average of **6 requests per minute**.

Question 1

What is the probability of exactly **4 requests in one minute**?

This is a **Poisson** problem.

$$X \sim \text{Poisson}(6)$$
$$P(X = 4) = \frac{e^{-6}6^4}{4!}$$

$$P(X = 4) \approx 0.1339$$

Therefore:

13.39%

Question 2

What is the probability that the **waiting time for the next request is less than 10 seconds**?

This is an **Exponential** problem.

Convert:

$$10 \text{ sec} = \frac{1}{6} \text{ minute}$$

Therefore:

$$P(X \leq \frac{1}{6}) = 1 - e^{-6(\frac{1}{6})}$$
$$P(X \leq \frac{1}{6}) = 1 - e^{-1}$$

$$P(X \leq 10 \text{ sec}) \approx 63.21\%$$



Poisson vs Exponential

This distinction is **very important in Data Science**.

Distribution	What does it measure?	Example
Poisson	Number of events	Number of website visitors per minute
Exponential	Waiting time between events	Time until next website visitor

Think of it like this:

Suppose customers arrive at a store.

Poisson asks:

"How many customers arrive in the next 10 minutes?"

Exponential asks:

"How long do I have to wait for the next customer?"

Therefore:

Poisson = COUNT

Exponential = WAITING TIME

Important Exponential Distribution Formulas

For:

$$X \sim \text{Exponential}(\lambda)$$

PDF

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

CDF

$$P(X \leq x) = 1 - e^{-\lambda x}$$

Survival probability

$$P(X > x) = e^{-\lambda x}$$

Interval probability

$$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$$

Mean

$$E(X) = \frac{1}{\lambda}$$

Variance

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

Standard deviation

$$\sigma = \frac{1}{\lambda}$$

Median

$$\text{Median} = \frac{\ln 2}{\lambda}$$

p-th percentile

$$x_p = \frac{-\ln(1-p)}{\lambda}$$



Memoryless Property

One of the most interesting properties of the Exponential distribution is the **memoryless property**.

It states:

$$P(X > s + t \mid X > s) = P(X > t)$$

In simple words:

How long you have already waited does not change the probability distribution of the additional waiting time.

Example

Suppose a server's failure time follows an Exponential distribution.

If it has already operated for 100 hours, the probability of surviving **another 10 hours** is the same as it would have been at the beginning, under the idealized Exponential model.

This property is mathematically useful, although real-world equipment often has aging or wear-out effects, so the Exponential assumption should be validated rather than automatically applied.



Exponential Distribution in Data Science

1. Web Analytics

Time until next visitor

2. API Monitoring

Time until next request

3. Queue Analytics

Time until next customer

4. Predictive Maintenance

Time until machine failure

5. Reliability Analysis

Component lifetime

6. Network Analytics

Time between network events

7. Customer Support

Time until next support request

8. Event-Driven Systems

Time between incoming events



Distribution Series — Quick Summary

You now have:

Distribution	Main Question	Data Science Example
Bernoulli	What happens in one trial?	Customer purchases or not
Binomial	How many successes in n trials?	Purchases among 100 customers
Geometric	How many trials until first success?	Customers until first purchase
Poisson	How many events occur in an interval?	API requests per minute
Uniform	Values equally spread across an interval	Random simulation value
Normal	How are values distributed around a mean?	Measurement/feature values
Exponential	How long until the next event?	Waiting time between API requests

The most important connection to remember:

Poisson → How many events?
Exponential → How much waiting time?

That distinction is **extremely useful when selecting probability distributions for real Data Science problems.**